

Adventist University of Central Africa

Master's in Big Data Analytics

Chapter 4 — Classification Lab

Introduction to Statistical Learning with Applications in R

Programme: MSc Big Data Analytics
Course Title: Data Mining
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1 Introduction

Linear regression predicts a *quantitative* response. But plenty of interesting questions have a *qualitative* answer instead — will the market go up or down today, will a customer buy the product or not, does this email look like spam. This lab works through the main tools for that kind of problem: logistic regression, linear and quadratic discriminant analysis (LDA/QDA), naive Bayes, and K -nearest neighbors, applied to three real data sets — daily S&P 500 returns, caravan insurance purchases, and hourly bike-share demand.

```
library(ISLR2)
library(MASS)
library(class)
library(e1071)
```

2 The Stock Market Data

`Smarket` records the percentage return of the S&P 500 on 1,250 trading days (2001–2005), along with the previous five days' returns (`Lag1` through `Lag5`), trading `Volume`, and `Direction` — whether the market went Up or Down that day. The question we're chasing: can *yesterday's* (and the last few days') returns predict *today's* direction at all?

```
dim(Smarket)
```

```
## [1] 1250    9
```

```
summary(Smarket)
```

##	Year	Lag1	Lag2	Lag3
##	Min. :2001	Min. :-4.922000	Min. :-4.922000	Min. :-4.922000
##	1st Qu.:2002	1st Qu.: -0.639500	1st Qu.: -0.639500	1st Qu.: -0.640000
##	Median :2003	Median : 0.039000	Median : 0.039000	Median : 0.038500
##	Mean :2003	Mean : 0.003834	Mean : 0.003919	Mean : 0.001716
##	3rd Qu.:2004	3rd Qu.: 0.596750	3rd Qu.: 0.596750	3rd Qu.: 0.596750
##	Max. :2005	Max. : 5.733000	Max. : 5.733000	Max. : 5.733000
##	Lag4	Lag5	Volume	Today
##	Min. :-4.922000	Min. :-4.92200	Min. :0.3561	Min. :-4.922000
##	1st Qu.: -0.640000	1st Qu.: -0.64000	1st Qu.:1.2574	1st Qu.: -0.639500
##	Median : 0.038500	Median : 0.03850	Median :1.4229	Median : 0.038500
##	Mean : 0.001636	Mean : 0.00561	Mean :1.4783	Mean : 0.003138
##	3rd Qu.: 0.596750	3rd Qu.: 0.59700	3rd Qu.:1.6417	3rd Qu.: 0.596750
##	Max. : 5.733000	Max. : 5.73300	Max. :3.1525	Max. : 5.733000
##	Direction			
##	Down:602			
##	Up :648			
##				
##				
##				
##				

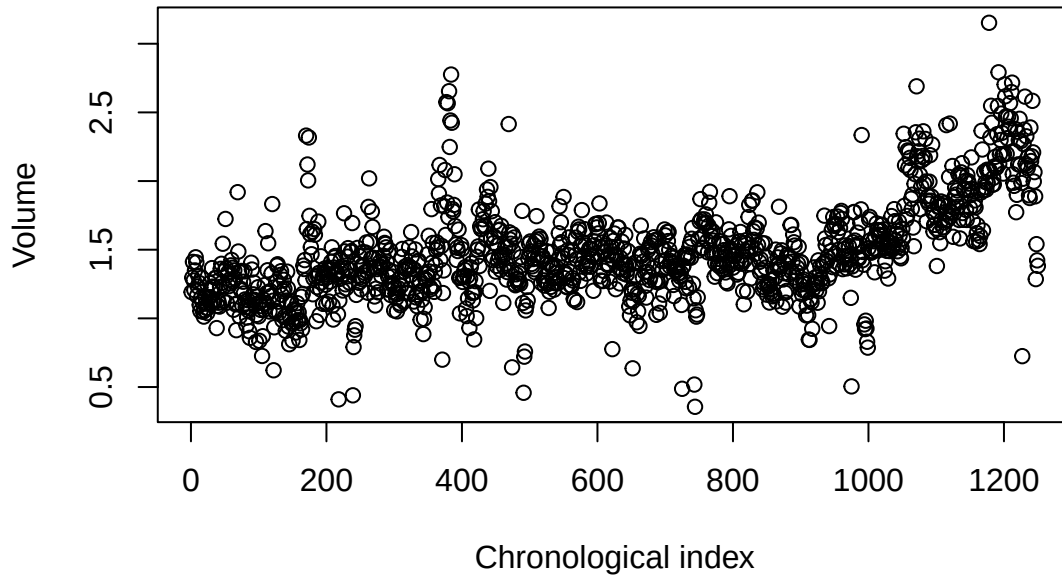
```
cor(Smarket[, -9])
```

```
##           Year      Lag1      Lag2      Lag3      Lag4
## Year  1.00000000  0.029699649  0.030596422  0.033194581  0.035688718
## Lag1  0.02969965  1.000000000 -0.026294328 -0.010803402 -0.002985911
## Lag2  0.03059642 -0.026294328  1.000000000 -0.025896670 -0.010853533
## Lag3  0.03319458 -0.010803402 -0.025896670  1.000000000 -0.024051036
## Lag4  0.03568872 -0.002985911 -0.010853533 -0.024051036  1.000000000
## Lag5  0.02978799 -0.005674606 -0.003557949 -0.018808338 -0.027083641
## Volume 0.53900647  0.040909908 -0.043383215 -0.041823686 -0.048414246
## Today 0.03009523 -0.026155045 -0.010250033 -0.002447647 -0.006899527
##           Lag5      Volume      Today
## Year  0.029787995  0.53900647  0.030095229
## Lag1 -0.005674606  0.04090991 -0.026155045
## Lag2 -0.003557949 -0.04338321 -0.010250033
## Lag3 -0.018808338 -0.04182369 -0.002447647
## Lag4 -0.027083641 -0.04841425 -0.006899527
## Lag5  1.000000000 -0.02200231 -0.034860083
## Volume -0.022002315  1.00000000  0.014591823
## Today -0.034860083  0.01459182  1.000000000
```

The correlations between the lag variables and Today's return all sit close to zero — there's essentially no linear relationship between recent past returns and today's. That's actually the expected, textbook result for daily stock returns (a market that were linearly predictable from its own recent past would be an unusually exploitable one). The only sizable correlation in the table is between Year and Volume — unsurprising, since trading volume trended upward over these five years:

```
plot(Smarket$Volume, xlab = "Chronological index", ylab = "Volume",
     main = "Trading volume increases over 2001-2005")
```

Trading volume increases over 2001–2005



3 Logistic Regression

Since `Direction` is binary, `lm()` isn't the right tool — nothing stops a linear model from predicting a probability below 0 or above 1. Logistic regression instead models the probability itself through the *logistic function*, which is mathematically guaranteed to stay in $(0, 1)$ no matter what the predictors are:

$$p(X) = P(\text{Direction} = \text{Up} \mid X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}$$

Rearranging shows *why* it's called “logistic” — it's linear not in the probability itself, but in the **log-odds**:

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

So a coefficient β_j is read as: a one-unit increase in X_j changes the log-odds of **Up** by β_j , holding everything else fixed — not a direct change in probability, since the logistic curve is flattest at the extremes and steepest near $p = 0.5$.

```
glm.fits <- glm(  
  Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,  
  data = Smarket, family = binomial  
)  
summary(glm.fits)
```

```
##  
## Call:  
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
```

```
##      Volume, family = binomial, data = Smarket)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.126000   0.240736  -0.523   0.601
## Lag1        -0.073074   0.050167  -1.457   0.145
## Lag2        -0.042301   0.050086  -0.845   0.398
## Lag3         0.011085   0.049939   0.222   0.824
## Lag4         0.009359   0.049974   0.187   0.851
## Lag5         0.010313   0.049511   0.208   0.835
## Volume       0.135441   0.158360   0.855   0.392
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1731.2  on 1249  degrees of freedom
## Residual deviance: 1727.6  on 1243  degrees of freedom
## AIC: 1741.6
##
## Number of Fisher Scoring iterations: 3
```

None of the p-values are small — by itself, this is a first sign that yesterday's returns don't carry much predictive signal for today's direction, consistent with what the correlation matrix already suggested. `Lag1` has the smallest p-value (≈ 0.145) with a *negative* coefficient, which would (weakly) suggest that a positive return yesterday makes a positive return today slightly less likely — but nowhere near significant on its own.

3.1 Confusion matrix and training accuracy

```
glm.probs <- predict(glm.fits, type = "response")
glm.pred <- rep("Down", nrow(Smarket))
glm.pred[glm.probs > 0.5] <- "Up"

table(glm.pred, Smarket$Direction)
```

```
##
## glm.pred Down  Up
##      Down  145 141
##      Up    457 507
```

```
mean(glm.pred == Smarket$Direction)
```

```
## [1] 0.5216
```

About 52% correct — barely better than a coin flip, and misleadingly so: this is **training** accuracy, computed on the exact data the model was fit to. A model that just memorizes noise in the training set can look fine here and still fail on new data — which is exactly why we need a genuine held-out test set.

3.2 A real train/test split

We use the classic “hold out the future” split for time series data: train on 2001–2004, and test on 2005 — data the model has never seen.

```
train <- Smarket$Year < 2005
Smarket.2005 <- Smarket[!train, ]
Direction.2005 <- Smarket$Direction[!train]
dim(Smarket.2005)

## [1] 252 9

glm.fits <- glm(
  Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
  data = Smarket, family = binomial, subset = train
)
glm.probs <- predict(glm.fits, Smarket.2005, type = "response")
glm.pred <- rep("Down", nrow(Smarket.2005))
glm.pred[glm.probs > 0.5] <- "Up"

table(glm.pred, Direction.2005)

##           Direction.2005
## glm.pred Down Up
##      Down   77 97
##      Up    34 44

mean(glm.pred == Direction.2005)

## [1] 0.4801587
```

Test accuracy is *worse* than training accuracy here — actually worse than just always guessing Up, which is a sign the full five-lag model is overfitting noise rather than finding real signal. Since Lag1 and Lag2 were the two most promising predictors above, it’s worth trying a leaner model:

```
glm.fits <- glm(Direction ~ Lag1 + Lag2, data = Smarket,
  family = binomial, subset = train)
glm.probs <- predict(glm.fits, Smarket.2005, type = "response")
glm.pred <- rep("Down", nrow(Smarket.2005))
glm.pred[glm.probs > 0.5] <- "Up"

table(glm.pred, Direction.2005)

##           Direction.2005
## glm.pred Down Up
##      Down   35 35
##      Up    76 106

mean(glm.pred == Direction.2005)

## [1] 0.5595238
```

That’s a meaningful improvement — dropping the three weakest predictors actually helped test

performance, a concrete illustration of the bias-variance idea from Chapter 2: a model with fewer, better-chosen predictors can generalize better than one that includes everything, even though it necessarily has strictly worse (or equal) training fit.

4 Linear Discriminant Analysis

Logistic regression models $P(Y | X)$ directly. LDA takes the opposite route — Bayes’ theorem in reverse — modeling how X is distributed *within* each class, then using Bayes’ rule to get back to $P(Y | X)$:

$$P(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_l \pi_l f_l(x)}$$

where π_k is the (estimated) prior probability of class k , and $f_k(x)$ is the density of X within class k . LDA’s specific assumption is that each class’s predictors follow a Gaussian distribution, all sharing the *same* covariance structure — only the class means differ. Under that assumption the decision boundary between classes works out to be linear, which is where LDA gets its name.

```
lda.fit <- lda(Direction ~ Lag1 + Lag2, data = Smarket, subset = train)
lda.fit
```

```
## Call:
## lda(Direction ~ Lag1 + Lag2, data = Smarket, subset = train)
##
## Prior probabilities of groups:
##      Down      Up
## 0.491984 0.508016
##
## Group means:
##           Lag1      Lag2
## Down 0.04279022 0.03389409
## Up   -0.03954635 -0.03132544
##
## Coefficients of linear discriminants:
##           LD1
## Lag1 -0.6420190
## Lag2 -0.5135293
```

The **prior probabilities** ($\hat{\pi}_{\text{Down}} \approx 0.492$, $\hat{\pi}_{\text{Up}} \approx 0.508$) are just the class proportions in the training data. The **group means** show that on Up days, Lag1 and Lag2 tend to be slightly negative on average — consistent with the (weak) negative association logistic regression also picked up.

```
lda.pred <- predict(lda.fit, Smarket.2005)
table(lda.pred$class, Direction.2005)
```

```
##      Direction.2005
##      Down  Up
## Down   35  35
## Up     76 106
```

```
mean(lda.pred$class == Direction.2005)
```

```
## [1] 0.5595238
```

Almost identical accuracy to the leaner logistic regression — not a coincidence. With two classes and (roughly) Gaussian, similarly-shaped predictors, LDA and logistic regression tend to land on very similar decision boundaries, even though they get there through different routes (generative vs. discriminative modeling).

5 Quadratic Discriminant Analysis

QDA relaxes LDA's shared-covariance assumption — each class gets its *own* covariance matrix. That extra flexibility means the decision boundary is now quadratic rather than linear, at the cost of estimating more parameters (and so needing more data to do it reliably).

```
qda.fit <- qda(Direction ~ Lag1 + Lag2, data = Smarket, subset = train)
qda.fit
```

```
## Call:
## qda(Direction ~ Lag1 + Lag2, data = Smarket, subset = train)
##
## Prior probabilities of groups:
##      Down      Up
## 0.491984 0.508016
##
## Group means:
##           Lag1      Lag2
## Down 0.04279022 0.03389409
## Up   -0.03954635 -0.03132544
```

```
qda.class <- predict(qda.fit, Smarket.2005)$class
table(qda.class, Direction.2005)
```

```
##           Direction.2005
## qda.class Down  Up
##      Down   30  20
##      Up    81 121
```

```
mean(qda.class == Direction.2005)
```

```
## [1] 0.5992063
```

QDA edges out both logistic regression and LDA here. One reading: the true boundary between **Up** and **Down** days may genuinely be closer to quadratic than linear — but with only two predictors and a few hundred test observations, this could also just be QDA getting fortunate on this particular test split rather than a guaranteed generalizable edge.

6 Naive Bayes

Naive Bayes keeps the same generative Bayes'-rule structure as LDA/QDA, but replaces the Gaussian-with-shared/class-specific-covariance assumption with something blunter and computationally cheaper: it assumes every predictor is **independent of every other predictor, within each class**.

$$f_k(x) = f_{k1}(x_1) \times f_{k2}(x_2) \times \cdots \times f_{kp}(x_p)$$

This is rarely *exactly* true — Lag1 and Lag2 are unlikely to be truly independent given Direction — but it collapses a hard multivariate density-estimation problem into p easy univariate ones, and in practice often still classifies well even when the independence assumption is technically wrong.

```
nb.fit <- naiveBayes(Direction ~ Lag1 + Lag2, data = Smarket, subset = train)
nb.fit
```

```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
##      Down      Up
## 0.491984 0.508016
##
## Conditional probabilities:
##      Lag1
## Y      [,1]      [,2]
## Down 0.04279022 1.227446
## Up   -0.03954635 1.231668
##
##      Lag2
## Y      [,1]      [,2]
## Down 0.03389409 1.239191
## Up   -0.03132544 1.220765
```

```
nb.class <- predict(nb.fit, Smarket.2005)
table(nb.class, Direction.2005)
```

```
##      Direction.2005
## nb.class Down  Up
##      Down   28  20
##      Up    83 121
```

```
mean(nb.class == Direction.2005)
```

```
## [1] 0.5912698
```

Performance here lands close to LDA and logistic regression — reasonable, since with only two predictors there’s less room for the independence assumption to cause real damage than there would be with many correlated predictors.

7 *K*-Nearest Neighbors

Every method so far assumes some parametric form for how `Direction` depends on the predictors (a linear log-odds, a Gaussian density, ...). KNN assumes nothing about the shape of the boundary at all: to classify a new point, find its *K* nearest neighbors in the training data (by Euclidean distance) and take a majority vote among them.

```
train.X <- cbind(Smarket$Lag1, Smarket$Lag2)[train, ]
test.X <- cbind(Smarket$Lag1, Smarket$Lag2)[!train, ]
train.Direction <- Smarket$Direction[train]

set.seed(1)
knn.pred <- knn(train.X, test.X, train.Direction, k = 1)
table(knn.pred, Direction.2005)
```

```
##           Direction.2005
## knn.pred Down Up
##      Down   43 58
##      Up    68 83
```

```
mean(knn.pred == Direction.2005)
```

```
## [1] 0.5
```

$K = 1$ is the most flexible possible choice — each test point is classified by whichever single training point happens to be closest, which chases noise aggressively and shows in the roughly 50% accuracy (no better than guessing). Bumping *K* up trades away some of that flexibility for stability:

```
knn.pred <- knn(train.X, test.X, train.Direction, k = 3)
table(knn.pred, Direction.2005)
```

```
##           Direction.2005
## knn.pred Down Up
##      Down   48 54
##      Up    63 87
```

```
mean(knn.pred == Direction.2005)
```

```
## [1] 0.5357143
```

A modest improvement, but still well short of QDA. Altogether, across every method tried, `Lag1/Lag2` simply don’t carry much signal about `Direction` — which, again, is the expected finding for daily stock returns, not a failure of the methods themselves.

7.1 Application: Caravan Insurance

`Caravan` is a more favorable setting for KNN: 5,822 customers, 85 demographic predictors, and a binary `Purchase` outcome (did they buy caravan insurance). Because KNN measures raw Euclidean

distance, any predictor on a larger numeric scale automatically dominates the distance calculation regardless of how informative it actually is — so predictors need to be standardized first:

```
standardized.X <- scale(Caravan[, -86])
mean(Caravan[[1]]); mean(Caravan[[2]])

## [1] 24.25335
## [1] 1.110615

var(standardized.X[, 1]); var(standardized.X[, 2])

## [1] 1
## [1] 1
```

After scaling, every predictor has variance 1, so no single variable can dominate the distance metric purely because of its units.

```
test <- 1:1000
train.X <- standardized.X[-test, ]
test.X <- standardized.X[test, ]
train.Y <- Caravan$Purchase[-test]
test.Y <- Caravan$Purchase[test]

set.seed(1)
knn.pred <- knn(train.X, test.X, train.Y, k = 3)
table(knn.pred, test.Y)

##           test.Y
## knn.pred  No  Yes
##       No   921  54
##       Yes   20   5
```

Only about 6% of customers actually purchase, so overall accuracy is a misleading metric here — a model that always predicts “No” would already be right about 94% of the time. What actually matters for a marketing campaign is *precision among predicted buyers*: of the customers KNN flagged as likely purchasers, what fraction really did buy? That’s a meaningfully higher hit rate than the 6% baseline rate, which is the actual value of the model — it concentrates marketing spend on customers substantially more likely to convert than a random customer would be.

8 Poisson Regression

Bikeshare records hourly bike rental counts (**bikers**) along with weather and calendar predictors. A count that’s always ≥ 0 and often small doesn’t fit the assumptions of ordinary linear regression well — a linear model can happily predict a negative number of bike rentals, which is nonsensical. Poisson regression is built for exactly this: it assumes $Y | X \sim \text{Poisson}(\lambda(X))$, and — just like logistic regression links probability through the logistic function — links the rate λ through a **log** link, guaranteeing predictions stay positive:

$$\log(\lambda(X)) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \quad \Longleftrightarrow \quad \lambda(X) = e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}$$

A coefficient β_j is therefore read multiplicatively: a one-unit increase in X_j multiplies the expected count by e^{β_j} , holding other predictors fixed.

```
mod.pois <- glm(
  bikers ~ mnth + hr + workingday + temp + weathersit,
  data = Bikeshare, family = poisson
)
summary(mod.pois)
```

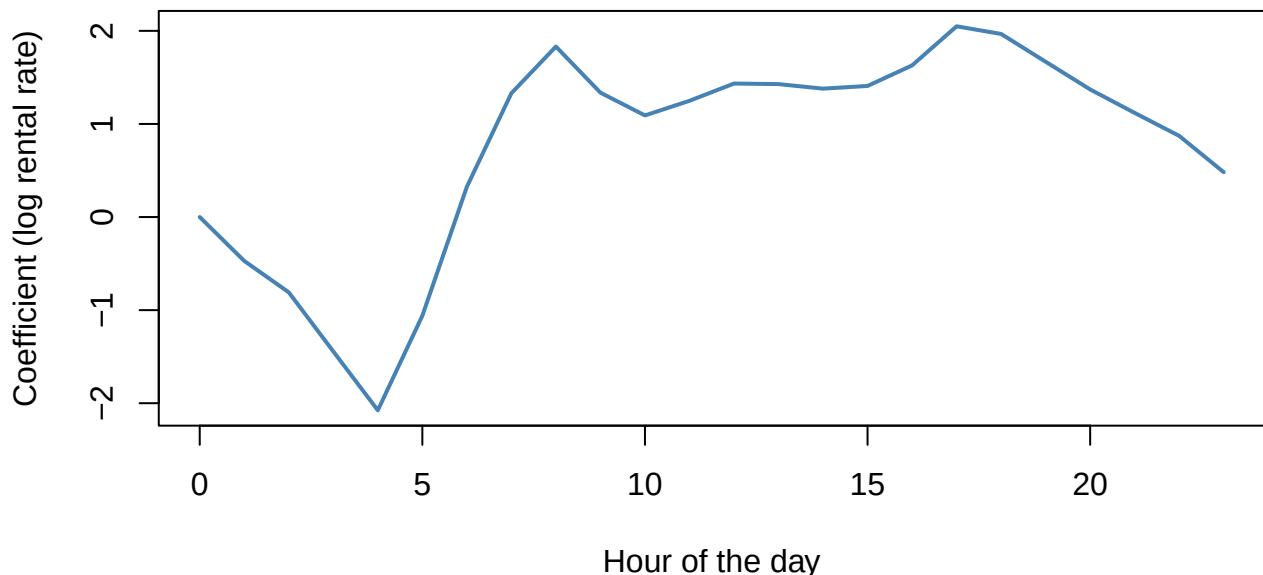
```
##
## Call:
## glm(formula = bikers ~ mnth + hr + workingday + temp + weathersit,
##      family = poisson, data = Bikeshare)
##
## Coefficients:
##
```

	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	2.693688	0.009720	277.124	< 2e-16 ***
## mnthFeb	0.226046	0.006951	32.521	< 2e-16 ***
## mnthMarch	0.376437	0.006691	56.263	< 2e-16 ***
## mnthApril	0.691693	0.006987	98.996	< 2e-16 ***
## mnthMay	0.910641	0.007436	122.469	< 2e-16 ***
## mnthJune	0.893405	0.008242	108.402	< 2e-16 ***
## mnthJuly	0.773787	0.008806	87.874	< 2e-16 ***
## mnthAug	0.821341	0.008332	98.573	< 2e-16 ***
## mnthSept	0.903663	0.007621	118.578	< 2e-16 ***
## mnthOct	0.937743	0.006744	139.054	< 2e-16 ***
## mnthNov	0.820433	0.006494	126.334	< 2e-16 ***
## mnthDec	0.686850	0.006317	108.724	< 2e-16 ***
## hr1	-0.471593	0.012999	-36.278	< 2e-16 ***
## hr2	-0.808761	0.014646	-55.220	< 2e-16 ***
## hr3	-1.443918	0.018843	-76.631	< 2e-16 ***
## hr4	-2.076098	0.024796	-83.728	< 2e-16 ***
## hr5	-1.060271	0.016075	-65.957	< 2e-16 ***
## hr6	0.324498	0.010610	30.585	< 2e-16 ***
## hr7	1.329567	0.009056	146.822	< 2e-16 ***
## hr8	1.831313	0.008653	211.630	< 2e-16 ***
## hr9	1.336155	0.009016	148.191	< 2e-16 ***
## hr10	1.091238	0.009261	117.831	< 2e-16 ***
## hr11	1.248507	0.009093	137.304	< 2e-16 ***
## hr12	1.434028	0.008936	160.486	< 2e-16 ***
## hr13	1.427951	0.008951	159.529	< 2e-16 ***
## hr14	1.379296	0.008999	153.266	< 2e-16 ***
## hr15	1.408149	0.008977	156.862	< 2e-16 ***
## hr16	1.628688	0.008805	184.979	< 2e-16 ***
## hr17	2.049021	0.008565	239.221	< 2e-16 ***
## hr18	1.966668	0.008586	229.065	< 2e-16 ***
## hr19	1.668409	0.008743	190.830	< 2e-16 ***
## hr20	1.370588	0.008973	152.737	< 2e-16 ***

```
## hr21          1.118568    0.009215   121.383 < 2e-16 ***
## hr22          0.871879    0.009536    91.429 < 2e-16 ***
## hr23          0.481387    0.010207    47.164 < 2e-16 ***
## workingday    0.014665    0.001955     7.502 6.27e-14 ***
## temp         0.785292    0.011475    68.434 < 2e-16 ***
## weathersitcloudy/misty -0.075231  0.002179   -34.528 < 2e-16 ***
## weathersitlight rain/snow -0.575800  0.004058  -141.905 < 2e-16 ***
## weathersitheavy rain/snow -0.926287  0.166782   -5.554 2.79e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 1052921  on 8644  degrees of freedom
## Residual deviance:  228041  on 8605  degrees of freedom
## AIC: 281159
##
## Number of Fisher Scoring iterations: 5
```

```
coef.hr <- c(0, coef(mod.pois)[grepl("^hr", names(coef(mod.pois)))])
plot(0:23, coef.hr, type = "l", pch = 19, col = "steelblue", lwd = 2,
     xlab = "Hour of the day", ylab = "Coefficient (log rental rate)",
     main = "Bikeshare demand by hour, holding other predictors fixed")
```

Bikeshare demand by hour, holding other predictors fixed



The hourly coefficients trace out the expected daily commuting rhythm — sharp peaks around the morning and evening rush hours, with a trough overnight — which a positive-count model like this one can represent naturally, in a way a plain linear regression fit to the raw counts would not.